ComfyUI + Wan 2.2: what everything actually does

Plain English. No definitions that define nothing. Every part and setting gets the same three answers.

What is it?
In words, not more acronyms.

What does it do?
To your video, or your wait.

What do I set it to?
A real starting value.

Sized for `16 GB VRAM` `32 GB RAM` with notes for smaller cards

1 The big picture READ THIS FIRST

Wan 2.2 does not film anything. It starts with a block of pure static, every frame at once, and cleans it up a little at a time until it looks like your prompt. The twist is that **two artists share the job**: one roughs out the layout and the motion, then hands over to a second one who finishes the detail.

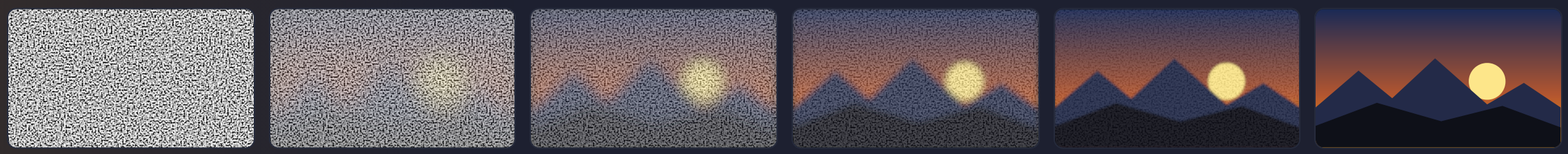
YOU
Prompt
Your words, plus a starting picture if you are doing image to video.

THE READER
Text encoder
Reads the prompt and turns it into numbers the artists can follow. Runs once per prompt.
`Load CLIP`

THE TWO ARTISTS
High noise, then low noise
The first takes the early steps: what is where, and how it moves. The second takes the late steps: faces, texture, sharpness. Nearly all of your wait is here.
`Load Diffusion Model, twice`

THE UNPACKER
VAE
The artists work on a small compressed version. The VAE turns it back into real frames.
`Load VAE`

RESULT
MP4
16 frames per second. No sound.



`start: static` `step 4` `step 8` `step 12` `step 16` `step 20: done`

A step is one cleanup pass. More passes, cleaner result, longer wait. (A drawing of the idea, not real output.)

Everything else on this page is one of three things: **a part** of that assembly line, **a file** that one of the parts loads, or **a knob** on the artists.

2 Versions WHICH WAN 2.2 YOU ACTUALLY WANT

"Wan 2.2" is a family, not one file. **Each job has its own model.** These four are the ones to know.

VERSION	YOU PLUG IN	WHAT IT DOES WITH IT	REACH FOR IT WHEN
T2V 14B Text to video	A prompt. Nothing else.	Invents everything: the people, the place, the camera.	You have no picture to start from. Needs the `t2v` pair of files.
I2V 14B Image to video	A prompt and one picture.	Your picture becomes the literal first frame. The video plays forward from it.	You already have the exact look you want and need it to move. Needs the `i2v` pair.
FLF2V First and last frame	A prompt and two pictures.	Opens on picture one, lands on picture two, and invents the trip between them.	The ending has to hit an exact image. It is not a separate download: it uses the `i2v` pair with a different node.
TI2V 5B The small one	A prompt, with or without a picture.	One single file that does both text to video and image to video, at 1280 x 704 and 24 frames per second.	Your card is small. The official docs say it fits well on 8 GB. It is the lightweight option, not the best-looking one.

Why the 14B models are two files

Each 14B model is really a pair: a `high_noise` file and a `low_noise` file. The high-noise one works while the frames are still mostly static and decides layout and motion. The low-noise one takes over once things have taken shape and refines detail. You need both. Only one is working at any moment.

Pairs do not mix

The `t2v` pair and the `i2v` pair are different models. A t2v file cannot take a starting picture, and high and low are not interchangeable either. If a result is noise or mush, a swapped file is the first thing to check.

The rest of the family

Each of these is its own download with its own template: **S2V** moves a character's mouth to a voice track you supply. **Animate** makes a character follow the movement in another video. **Fun Control, Inpaint, Camera and VACE** steer a video with poses, masks or camera paths.

3 The parts WHAT EACH FILE YOU LOAD IS FOR

A 14B workflow loads **four model files**: two artists, one reader, one unpacker.

Diffusion models THE TWO ARTISTS

`also called: checkpoint, UNet, DiT | field: unet_name`

WHAT THEY DO

Make the video. Each is about 14 billion numbers. Together that is 27 billion, but only one of them is in use on any given step.

WHY YOU CARE

They are the big files and they do the work on every step. Halfway through a run, ComfyUI puts the first one away and brings out the second, and that swap is a real pause on a mid-range PC.

Text encoder THE READER

`also called: CLIP | field: clip_name`

WHAT IT DOES

Reads your prompt and hands the artists numbers that mean the same thing. It reads text only. It never sees your starting picture.

WHY YOU CARE

It is a language model called umT5-XXL, and the same file serves every Wan 2.2 version. It runs once per prompt, for the positive and the negative prompt. Change only the seed and ComfyUI reuses its last answer.

Set the loader's type to `wan`.

VAE THE UNPACKER

`field: vae_name`

WHAT IT DOES

Turns the artists' compressed result into real frames, and squeezes your starting picture down on the way in.

WHY YOU CARE

There are two and they do not swap. The 14B models use `wan_2.1_vae`. Yes, the 2.1 one: that is not a typo. Only the 5B model uses `wan2.2_vae`. Unpacking the whole clip is also a burst of memory at the very end.

Add-ons and words you will keep seeing

LoRA

A small patch file that sits on top of an artist and bends it toward something: a face, a style, a kind of motion, or speed. **Strength** is how hard it bends. 0 is off, 1.0 is what its maker intended. For the 14B models, LoRAs usually come as a **pair**, one for each artist. Load the high one on the high-noise model and the low one on the low-noise model.

Lightning LoRA LIGHTX2V, 4 STEPS

A pair of LoRAs that teach the artists to finish in 4 steps where they normally take 20. It is the single biggest speed-up there is for this model. The official templates have it built in behind one switch. The price is in section 5.

Checkpoint

Just means a saved model file. People use it loosely for the diffusion model.

.safetensors

The container, like .zip or .mp4. It tells you nothing about what is inside. The name is literal: the file can only hold numbers, so it cannot hide a program the way the older .ckpt and .pt files could.

4 File names WHICH ONE TO DOWNLOAD

Read a model's filename left to right and it tells you exactly what you are getting.

`wan2.2` `_t2v` `_high_noise` `_14B` `_fp8_scaled` `.safetensors`

wan2.2
The family and version. Wan 2.1 files are a different model.

t2v
The job. `t2v` text, `i2v` image, `ti2v` both, `s2v` voice, `animate`, `fun_...`

high_noise
Which artist: the first (`high`) or the second (`low`). You need both.

14B
Size, in billions of numbers. 14B is the full model, 5B the small one.

fp8_scaled
How its numbers are stored. This sets the file size.

.safetensors
The container. Ignore it.

The one idea behind all the letters

A model is a giant list of numbers. The letters say **how many bits each number is stored with**. Fewer bits: smaller file, more of it fits on your card, slightly less exact result. Same trade as JPEG quality or MP3 bitrate.

fp16

16 bits per number. The original, full-quality file, and the biggest one. You will also see bf16, which is the same idea with the 16 bits laid out differently.

fp8_scaled

8 bits per number, with a scale factor kept for each layer so less is lost in the rounding. Half the size of the 16-bit original. It is the long-standing 8-bit format and runs on nearly any setup.

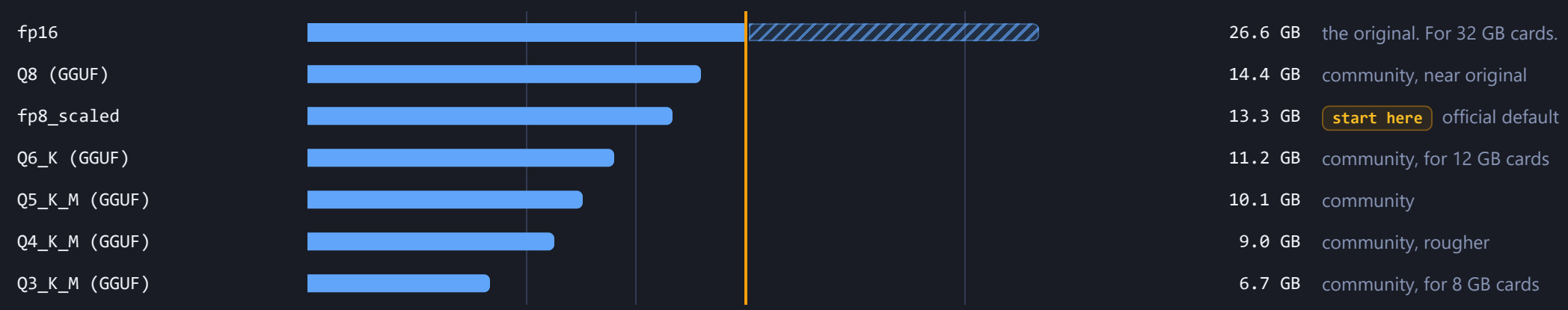
GGUF (Q8, Q6, Q5, Q4 ...)

A community packaging format borrowed from the chatbot world. The Q number is roughly the bits per number: Q8 is close to the original, Q4 is about half that size and rougher. It needs the ComfyUI-GGUF custom node pack and its own loader node, and the files go in `models/unet`.

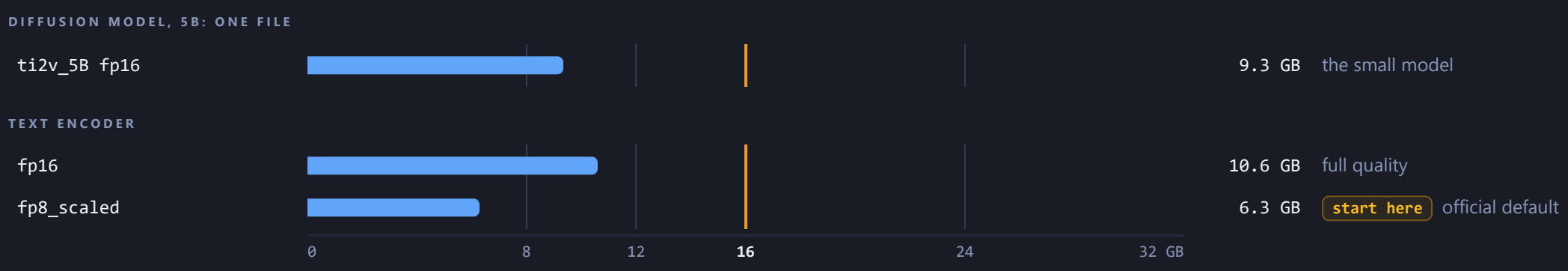
File size against your card

fits on a 16 GB card · past 16 GB, spills into system RAM · 16 GB · faint lines: 8, 12 and 24 GB

DIFFUSION MODEL, 14B: SIZE OF EACH OF THE TWO FILES (T2V SHOWN, I2V IS THE SAME)



DIFFUSION MODEL, 5B: ONE FILE



Official sizes are from the Comfy-Org files. GGUF sizes are from one community set (QuantStack, t2v). Video needs a lot of working room on the card on top of the file itself, so a file that sits right under your line will still spill a little.

Download this set (text to video)

PART	FILE	SIZE	GOES IN COMFYUI/MODELS/
Diffusion model the first artist	wan2.2_t2v_high_noise_14B_fp8_scaled.safetensors	13.3 GB	diffusion_models
Diffusion model the second artist	wan2.2_t2v_low_noise_14B_fp8_scaled.safetensors	13.3 GB	diffusion_models
Text encoder	umt5_xxl_fp8_e4m3fn_scaled.safetensors	6.3 GB	text_encoders
VAE	wan_2.1_vae.safetensors	0.2 GB	vae
Lightning LoRAs optional, both files	wan2.2_t2v_lightx2v_4steps_lora_v1.1_high_noise.safetensors wan2.2_t2v_lightx2v_4steps_lora_v1.1_low_noise.safetensors	1.1 GB 1.1 GB	loras

For image to video, swap the two diffusion models for `wan2.2_i2v_high_noise_14B_fp8_scaled` and `wan2.2_i2v_low_noise_14B_fp8_scaled`, and the LoRAs for the `i2v_lightx2v_4steps` pair. The text encoder and VAE stay the same.

These are the exact files ComfyUI's built-in Wan 2.2 templates ask for. All of them are at huggingface.co/Comfy-Org/Wan_2.2_ComfyUI_Repackaged, inside the `split_files` folder. Wan 2.2 is under the Apache 2.0 license.

Smaller is not automatically better. A smaller file helps for one reason: less of it spills off the card. Every step down in bits costs a little exactness, so go only as small as you have to.

5 Settings THE KNOBS ON THE ARTISTS

Change **one at a time, with the seed fixed**, or you will never know which one did it. Values are for the 14B models. The 5B model's own defaults are underneath.

KNOB	WHAT IT ACTUALLY DOES	TURN IT UP AND	START AT
Resolution width x height	How many pixels are in each frame.	Frames get sharper. Every step takes longer and needs more memory, climbing at least as fast as the pixel count. The 14B models are built for 480p and 720p, and 720p has more than twice the pixels of 480p.	`640 x 640` The template default, chosen for smaller cards. Then 832 x 480, then 1280 x 720 if your card can take it.
Length frames, or duration in seconds	How long the clip is. Frames = seconds x 16, plus 1. The count moves in steps of 4, so 81 frames is 5 seconds.	Longer clip, and more work and memory on every single step. The official workflow is built around 5 seconds.	`81 frames = 5 seconds` Drop to 49 (3 seconds) if you are short on memory.
FPS frames per second	Not a knob. The 14B models make 16 frames for every second. The 5B model makes 24.	If you change it in the save node, the same frames just play faster or slower.	`16` Leave it. Smoothing to a higher frame rate is a separate step done afterwards.

Seed	The exact pattern of static the artists start from. It is a number that picks a pattern, nothing more.	There is no up or down, only different. Same seed with everything else the same gives you the same video again. A new seed is a new take on the same prompt.	`fixed`
			While you test a setting.
			`random`
			When you just want more takes.

Steps	How many cleanup passes the clip gets in total, shared between the two artists.	Cleaner picture and steadier motion. Time rises in a straight line.	`20` The official default without the Lightning LoRA.
Split step the hand-over	The step where the first artist stops and the second takes over. In the node it shows up as an end step on one sampler and a start step on the other.	The first artist keeps the clip longer: more of the run goes to layout and motion, less to detail. Lower it and the reverse happens.	`10 of 20` Half way. Leave it there.
CFG	How hard to push toward your prompt and away from the negative prompt. Above 1, the artist runs twice on every step, once for each.	The video follows the prompt harder. Past a point it overcooks: harsh contrast, burnt colour, jittery motion.	`3.5` The official value. It is why 20 steps here really means 40 passes.
Negative prompt	A list of things you do not want. Unlike some newer models, it works here, because CFG is above 1.	Does not apply. The template ships a stock list in Chinese that asks for no blur, no subtitles, no extra fingers, no frozen frames.	`leave the stock one` Add to it if something keeps showing up.
Shift	Where the steps get spent. Higher puts more of them on the early, blurry stage where composition and movement are decided. Lower puts more on late fine detail.	The balance tips toward structure and away from detail.	`5` The value in the official text and image to video templates. The first and last frame template uses 8.
Sampler	The method the artists use on each pass to decide what to clean up next.	Not a more or less knob. A different sampler reaches a slightly different result in the same number of steps.	`euler` The official pick. Do not tune this early.
Scheduler	The pacing. It decides how much static comes off at each step: big chunks early, fine touches late.	Not a more or less knob either. Different pacing, same number of steps.	`simple` Leave it.

Lightning LoRA
one switch in the templates
	Loads the 4-step LoRA pair and changes three things at once: steps to 4, split step to 2, CFG to 1.	Not a dial, a switch. On, a run is several times faster: the official notes list about 9 minutes dropping to under 2 on a 24 GB card at 640 x 640. The price, also from the official notes, is less motion. And with CFG at 1 the negative prompt is ignored.	`on for drafts`
			`off for keepers`
			When motion matters most.

LoRA strength	How hard a LoRA bends an artist.	Stronger effect, then overcooked: warped shapes, flicker, a look that swallows the prompt.	`1.0` Then nudge by 0.1 or 0.2. Keep the high and low halves of a pair the same.

The 5B model has its own defaults

RESOLUTION	LENGTH	FPS	STEPS	CFG	SHIFT	SAMPLER	VAE
1280 x 704	121 frames = 5 s	24	20	5	8	uni_pc, simple	wan2.2_vae

One file, one sampler, no hand-over. These are the values in ComfyUI's built-in 5B template.

6 Your hardware WHY IT IS SLOW, AND WHAT THE ERRORS MEAN

The 14B models are heavy, mostly because there are two of everything. They run on a 16 GB card. They are not quick.

VRAM THE WORKBENCH

The graphics card's own memory. Whatever is sitting on it gets worked on at full speed, so this is the number that decides what is comfortable. This guide assumes 16 GB.

RAM THE SHELF

The PC's regular memory. Model files wait here when it is not their turn, and anything that does not fit on the workbench lives here. This guide assumes 32 GB.

Offloading THE SHUFFLE

When a model is bigger than the workbench, ComfyUI keeps the overflow on the shelf and swaps pieces on and off the card as each is needed. It is automatic. The result comes out the same. It only costs time.

The numbers

The default set is about **33 GB of model**: two artists at 13.3 each, reader 6.3, unpacker 0.2. They **take turns**, and only one artist is on the card at a time. A 13.3 GB artist is under the 16 GB line, but video needs a lot of working room on top, so expect part of it to spill. Half way through, the first artist is put away and the second is brought out. With 32 GB of RAM the pair only just fits on the shelf beside everything else, so that swap can end up reading from the disk. **Keep the models on an SSD.**

Too slow, or out of memory? Pull these in order

1. **Turn the Lightning LoRA on.** The biggest time saving there is. It does not save memory.
2. **Smaller frame.** 640 x 640 or 832 x 480.
3. **Shorter clip.** 49 frames is 3 seconds.
4. **Smaller files.** A GGUF pair, Q5 or Q4, with the ComfyUI-GGUF node pack.
5. **The 5B model.** One 9.3 GB file in place of two 13.3 GB ones.
6. **Free the machine.** Close everything else that uses RAM or the graphics card.

What an error is telling you

`Red boxes when a workflow opens`
Missing custom nodes, or the workflow needs a newer ComfyUI. Update first, then let ComfyUI Manager install what is missing.

`"Value not in list" on a loader`
That file is not in the right models folder, or it has a different name than the workflow expects. Pick yours from the dropdown.

`"CUDA out of memory" / "Allocation on device"`
The card ran out of VRAM. Work down the list on the left.

`Dies after the last step finished`
Usually the VAE running out while unpacking. Lower the resolution.

`The whole PC crawls or freezes`
RAM is full and Windows is using the disk as memory. Close other programs, or move to smaller files.

The other words in this corner

CUDA / PyTorch THE PLUMBING

PyTorch is the math engine ComfyUI is built on. CUDA is NVIDIA's layer that lets PyTorch use the graphics card. Both come bundled with ComfyUI and you never touch them. You only meet them in messages: "CUDA out of memory" means the card is full, and a tag like `cu130` in the startup log is just the CUDA version PyTorch was built for.

Custom node AN ADD-ON

Every box on the canvas is a node. Custom nodes are boxes written by other people, installed through ComfyUI Manager. **Wan 2.2 itself needs none**: its nodes are built into ComfyUI. The one exception in this guide is GGUF files, which need the ComfyUI-GGUF pack. You only need a pack when a workflow you downloaded uses one. Each pack is code running on your PC, so stick to well-known ones.

rgthree A POPULAR PACK

A set of convenience nodes that shows up in a lot of shared workflows. The two you will meet most: a LoRA loader that stacks several LoRAs in one box, and switches that turn whole groups of a workflow on and off.

7 First run A KNOWN-GOOD STARTING POINT

Get one boring clip working with the official setup before you touch anything clever.

The recipe

1. **Update ComfyUI.** The templates change often.
2. **Open the built-in template.** Workflow menu, Browse Templates, search for Wan 2.2, pick the 14B text to video one.
3. **Get the files.** ComfyUI lists what is missing and where it goes. It is the set from section 4, LoRAs included.
4. **First test: Lightning on.** 640 x 640, 5 seconds. This proves everything loads without a long wait.
5. **Second test: Lightning off.** Same seed, same prompt. It will take several times longer. Compare the motion.
6. **Then change one thing.** Fix the seed, change a single knob, compare.

The prompt

Describe it like a shot. Who or what, doing what, where, in what light, and how the camera moves: "the camera slowly pushes in", "a slow pan to the left". Long and specific beats short and vague.

For image to video, the picture already shows what things look like. Spend the prompt on what happens next and what the camera does.

There is no sound, so dialogue in the prompt gives you moving lips at best, not a voice.

Leave the negative prompt alone until you have a reason to change it. It only works with the Lightning LoRA off.